

**AN IMPROVED HIERARCHICAL DEEP
REINFORCEMENT LEARNING FOR COMPLEX
IMPERFECT-INFORMATION CARD GAMES**

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一种用于复杂不完全信息卡牌游戏的改进型分层深度强化学习算法

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马来西亚 SAINS 大学

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by

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一种用于复杂不完全信息卡牌游戏的改进型分层深度强化学习算法

通过

罗倩

为满足相关要求而提交的学位论文
关于.....的程度
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- Appendix A Method Design and Implementation
- Appendix B Experiment Results and Discussion

附录清单

附录 A: 方法设计与实现 附录 B: 实验结果与讨论

**PEMBELAJARAN PENGUKUHAN MENDALAM HIERARKI YANG
DIPERBAIKI UNTUK PERMAINAN KAD MAKLUMAT TIDAK
SEMPURNA DAN KOMPLEKS**

ABSTRAK

Pembelajaran pengukuhan mendalam (DRL) telah mencapai kejayaan besar dalam pelbagai permainan, sama ada dengan maklumat sempurna atau tidak sempurna, seperti Go, Texas Hold'em, dan StarCraft II. Walau bagaimanapun, DouDiZhu dan Big2, permainan kad klasik yang kompleks dan maklumat tidak sempurna yang sangat popular di Asia, mempersembahkan cabaran baru bagi Kecerdasan Buatan (AI) dari segi persaingan, kerjasama, meneka maklumat yang tidak sempurna, mengendalikan ruang tindakan yang besar, dan latihan dengan ganjaran yang jarang. Kaedah Deep Monte Carlo (DMC) untuk permainan kad ini mencapai kejayaan yang signifikan tetapi masih menghadapi tiga masalah penyelidikan utama: kelajuan pembelajaran yang perlahan, kerugian yang tinggi semasa pembelajaran, dan pengoptimuman prestasi. Objektif utama penyelidikan ini adalah untuk meningkatkan prestasi AI untuk permainan kad maklumat tidak sempurna yang kompleks ini dengan rangka kerja Pembelajaran Pengukuhan Mendalam Hierarki (HDRL). Khususnya, matlamat utama ini dibahagikan kepada tiga objektif penyelidikan kecil: meningkatkan kecekapan pembelajaran dalam latihan DMC melalui Pembimbingan Oracle Guiding, meningkatkan kestabilan pembelajaran dengan Monte Carlo Mendalam Adaptif (ADMC), dan meningkatkan prestasi Pengoptimuman Dasar Proksimal (PPO) menggunakan Pembentukan Ganjaran Kelebihan Relatif (RARS). Rangka kerja HDRL mengintegrasikan Oracle Guiding, ADMC, RARS, dan PPO, membentuk proses pembuatan keputusan dua peringkat. Dengan mengabstraksi dan mempermudah ruang tindakan yang besar, model dasar

基于 UNTUK 、 KAD 及 TIDAK 信息的分层深度验证教学

简而繁

ABSTRAK

深度强化学习（DRL）已在多种博弈场景中取得显著成就，无论面对完美信息还是非完美信息环境，例如围棋、德州扑克及《星际争霸II》。然而，在亚洲广受欢迎的复杂非完美信息卡牌游戏《斗地主》与《Big2》中，人工智能（AI）面临着全新挑战：竞争对抗、协作博弈、基于非完美信息的信息推断、庞大行动空间的操控，以及稀疏奖励下的训练问题。针对此类卡牌游戏，深度蒙特卡洛（DMC）方法已取得显著进展，但仍面临三大核心研究难题：学习速度缓慢、训练期间损失较高，以及性能优化不足。本研究的核心目标在于采用分层深度强化学习（HDRL）框架，提升AI在复杂非完美信息卡牌游戏中的性能表现。具体而言，这一主要目标被分解为三个小型研究目标：通过Oracle Guiding提升DMC训练中的学习效率；借助自适应深度蒙特卡洛方法（ADMC）增强学习稳定性；以及利用相对优势奖励生成机制（RARS）提升近端策略优化（PPO）的性能。HDRL框架整合了Oracle Guiding、ADMC、RARS及PPO，构建出双层决策流程。

通过对广阔行动空间进行抽象化与简化，采用PPO与RARS训练的基模型可做出

utama, yang dilatih dengan PPO dan RARS, membuat keputusan strategik peringkat tinggi, yang selanjutnya diperhalusi menjadi tindakan khusus oleh model sub-nilai, yang dilatih dengan Oracle Guiding dan ADMC. Oracle Guiding meningkatkan kecekapan pembelajaran dengan melatih agen Oracle menggunakan kedua-dua maklumat tidak sempurna dan sempurna, kemudian secara beransur-ansur mengurangkan pergantungan pada maklumat yang sempurna. Eksperimen ablasi menunjukkan bahawa Oracle Guiding belajar lebih cepat daripada kaedah asal dalam tempoh latihan yang sama. ADMC meningkatkan kestabilan pembelajaran dengan menggunakan klip berat kecerunan untuk mengehadkan magnitud kemas kini, dengan itu mencegah perubahan polisi yang melampau. Hasil eksperimen menunjukkan bahawa ADMC mencapai kestabilan yang lebih baik, dengan kerugian yang lebih kecil dan kurang turun naik, berbanding kaedah asal di bawah keadaan yang sama. RARS menjana isyarat ganjaran untuk Penganggar Kelebihan Umum (GAE) dalam PPO, membolehkan latihan yang efektif dalam persekitaran ganjaran yang jarang. Hasil eksperimen menunjukkan bahawa RARS meningkatkan prestasi PPO dalam permainan kad kompleks ini dengan menyediakan ganjaran yang lebih padat. Hasil penilaian menunjukkan bahawa rangka kerja HDRL meningkatkan prestasi AI secara signifikan dalam kedua-dua DouDiZhu dan Big2. Eksperimen perbandingan dengan garis dasar terkini menunjukkan bahawa rangka kerja HDRL mengatasi model DMC terkini selepas 30 hari latihan sendiri. Penemuan ini mengesahkan keberkesanan kaedah yang dicadangkan dalam meningkatkan kecekapan pembelajaran, kestabilan, dan prestasi dalam permainan kad maklumat yang tidak sempurna yang kompleks.

高层战略决策；这些决策随后由采用Oracle Guiding与ADMC训练的次级模型细化为具体行动。Oracle Guiding通过让Oracle代理同时使用不完整与完整信息进行训练，进而逐步降低对完整信息的依赖，从而提升学习效率。消融实验表明，在相同训练周期内，Oracle Guiding的学习速度优于原始方法。ADMC则采用斜率裁剪技术限制更新幅度，从而抑制过度政策调整，进而提升ke-stabilanpembelajaran dengan性能。实验结果表明：在相同条件下，ADMC相较于原始方法展现出更优的稳定性——损失更小且波动更缓。RARS在PPO框架中为通用优势评估器（GAE）生成奖励信号，使得在稀疏奖励环境中仍能实现有效训练。实验结果表明，RARS通过提供更密集的奖励机制，提升了PPO模型在此复杂卡牌游戏中的性能。评估结果显示，HDRL框架在DouDiZhu与Big2两项任务中均显著提升了AI模型的性能。与最新基准模型的对比实验表明，HDRL框架在自主训练30天后，其性能已超越最新DMC模型。这些发现进一步证实：所提出的方法能有效提升在复杂、不完美信息卡牌游戏中的学习效率、稳定性及整体性能。

AN IMPROVED HIERARCHICAL DEEP REINFORCEMENT LEARNING FOR COMPLEX IMPERFECT-INFORMATION CARD GAMES

ABSTRACT

Deep Reinforcement Learning (DRL) has achieved significant breakthroughs in a variety of games, both with perfect and imperfect information, such as Go, Texas Hold'em, and StarCraft II. However, DouDiZhu and Big2 are classic complex card games with imperfect information and are popular in Asia. They present new challenges for AI in competition, cooperation, inferring imperfect information, handling large state-action spaces, and training with sparse rewards. The Deep Monte Carlo (DMC) method for these card games achieves significant success but still faces three key research problems: slow learning speed, high loss during learning, and performance optimization. The primary objective of this research is to enhance the performance for these complex imperfect-information card games with a Hierarchical Deep Reinforcement Learning (HDRL) framework. Specifically, this main goal is divided into three sub-research objectives: improving learning efficiency in DMC training through Oracle Guiding, enhancing learning stability with Adaptive Deep Monte Carlo (ADMC), and improving the performance of Proximal Policy Optimization (PPO) using Relative Advantage Reward Shaping (RARS). The HDRL framework integrates Oracle Guiding, ADMC, RARS, and PPO, forming a two-stage decision-making process. By abstracting and simplifying the large action space, the main-policy model, trained with PPO and RARS, makes high-level strategic decisions, which are further refined into specific actions by the sub-value model, trained with Oracle Guiding and ADMC. Oracle Guiding improves learning efficiency by training an Oracle agent using both imperfect and perfect information, then gradually reducing reliance on the former. Ablation experi-

一种改进的分层深度强化学习算法

适用于复杂的不完全信息卡牌游戏

摘要

深度强化学习（DRL）已在多种游戏中取得了重大突破，无论是完全信息还是不完全信息下的游戏，如围棋、德州扑克和星际争霸II。然而，斗地主和大2是经典的复杂卡牌游戏，具有不完全信息特性，在亚洲广受欢迎。这些游戏为人工智能带来了新的挑战，包括竞争与协作、推断不完全信息、处理庞大的状态-动作空间以及使用稀疏奖励进行训练。针对这些卡牌游戏，深度蒙特卡洛（DMC）方法已取得显著成果，但仍面临三个关键研究问题：学习速度缓慢、学习过程中损失较高以及性能优化不足。本研究的主要目标是利用分层深度强化学习（HDRL）框架，提升这些复杂不完全信息卡牌游戏的性能。具体而言，这一主要目标分为三个子研究目标：通过“预言机引导”提高DMC训练中的学习效率；利用“自适应深度蒙特卡洛”（ADMC）增强学习稳定性；以及通过“相对优势奖励塑造”（RARS）优化近端策略优化（PPO）的性能。HDRL框架整合了 Oracle Guiding、ADMC、RARS 和 PPO，构成了一个两阶段决策流程。通过抽象并简化庞大的动作空间，主策略模型——使用 PPO 和 RARS 训练——负责制定高层战略决策，而这些决策随后由次级价值模型——使用 Oracle Guiding 和 ADCMC 训练——细化为具体动作。Oracle Guiding 通过使用不完全信息和完全信息训练 Oracle 智能体，然后逐步减少对前者的依赖，从而提

ments demonstrate that Oracle Guiding learns faster than the original method within the same training period. ADMC enhances learning stability by employing gradient weight clipping to constrain the magnitude of updates, thereby preventing extreme policy changes. Experiment results show that ADMC achieves better stability, with smaller loss and less fluctuation, than the original method under identical conditions. RARS generates reward signals for the Generalized Advantage Estimator (GAE) in PPO, enabling effective training in sparse reward environments. Experiment results show that RARS significantly improves the performance of PPO in these complex card games by providing denser rewards. Evaluation results demonstrate that the HDRL framework significantly improves AI performance in both DouDiZhu and Big2. Comparative experiments with state-of-the-art baselines show that the HDRL framework outperforms the state-of-the-art DMC model after 30 days of self-play and training. These findings confirm the effectiveness of the proposed methods in enhancing learning efficiency, stability, and performance in complex imperfect-information card games.

高了学习效率。消融实验表明，在相同训练周期内，Oracle Guiding 的学习速度比原始方法更快。ADMC 则通过采用梯度权重裁剪来限制更新幅度，从而防止策略发生剧烈变化，进而增强了学习稳定性。实验结果表明，在相同条件下，ADMC 相比原始方法具有更好的稳定性，其损失更小，波动也更小。

RARS 为 PPO 中的广义优势估计器（GAE）生成奖励信号，从而在稀疏奖励环境中实现有效训练。实验结果表明，RARS 通过提供更密集的奖励，显著提升了 PPO 在这些复杂卡牌游戏中的性能。评估结果证明，HDRL 框架在 DouDiZhu 和 Big2 中均显著提升了人工智能的性能。与当前最先进基线模型的对比实验显示，经过 30 天的自玩和训练后，HDRL 框架的表现优于最先进的 DMC 模型。

这些研究结果证实，所提出的方法在提升复杂不完全信息卡牌游戏的学习效率、稳定性和性能方面具有显著效果。

CHAPTER 1

INTRODUCTION

Deep Reinforcement Learning (DRL) (Sutton and Barto, 2018; Wang et al., 2022b; Yin et al., 2024) has achieved several breakthroughs in game playing. In particular, perfect information games, where the complete game state is accessible to all players, have seen machines outperforming human professional players. Examples include Atari (Mnih et al., 2015), Chess (Silver et al., 2017), Go (Schrittwieser et al., 2020), and Tic-Tac-Toe (van Opheusden et al., 2023). However, in some imperfect information games, where some information is hidden from players, humans still have an advantage over state-of-the-art agents. Recent advances, though, show that this gap is narrowing, as seen in real-time strategy game StarCraft II (Vinyals et al., 2019) and board game Stratego (Perolat et al., 2022).

Complex imperfect-information card games are a subset of imperfect information games. They are valuable not only because they attract millions of players daily but also because they reflect numerous real-world scenarios, such as negotiations, surgical operations, business, and physics. Games provide a virtual world for Artificial Intelligence (AI) to explore, mirroring real-life problems like real-time decision making, scheduling, and education (Hu et al., 2024). Such games serve as perfect testbeds to verify the ability, generality, robustness, and safety of AI. This allows the methods and strategies refined through the study of these card games to be applied effectively in various practical contexts.

第一章 引言

深度强化学习（DRL）（Sutton 与 Barto, 2018; Wang 等人, 2022b; Yin 等人, 2024）在博弈领域已取得多项突破。尤其在完美信息博弈中——即所有玩家均可获取完整游戏状态——机器已超越人类职业玩家，典型案例包括 Atari（Mnih 等人, 2015）、国际象棋（Silver 等人, 2017）、围棋（Schrittwieser 等人, 2020）以及井字棋（van Opheusden 等人, 2023）。然而，在某些不完全信息博弈中——即部分信息对玩家不可见——人类仍较当前最先进的智能体更具优势。但近期进展表明，这一差距正在缩小，例如在实时策略游戏《星际争霸II》（Vinyals 等人, 2019）与棋盘游戏《Stratego》（Perolat 等人, 2022）中均可见一斑。

复杂不完全信息卡牌游戏是不完全信息游戏的一个子类。这类游戏之所以具有重要价值，不仅在于每日吸引数百万玩家参与，更在于它们能够映射出众多现实世界场景——例如谈判、外科手术、商业活动及物理现象等。这些游戏为人工智能（AI）提供了探索的虚拟环境，可模拟实时决策、任务调度及教育等现实问题（Hu 等人, 2024）。此类游戏可作为理想测试平台，用以验证人工智能的能力、普适性、鲁棒性及安全性。由此，通过研究这些卡牌游戏而优化的方法与策略，便可有效应用于各类实际场景。

Despite the potential, training state-of-the-art models for complex card games like DouDiZhu and Big2 remains challenging due to their inherent characteristics, including imperfect information with multiple players, a large and complex state and action space, and sparse rewards. This research aims to enhance the performance for these games through a Hierarchical Deep Reinforcement Learning (HDRL) framework. HDRL is a DRL method that decomposes complex tasks into multiple sub-tasks and uses a hierarchical structure for learning and decision-making (Chai et al., 2023; Gao et al., 2024; Vezhnevets et al., 2017). By breaking down intricate tasks into a series of simpler sub-tasks, with each layer focusing on solving decision problems at a specific level, it enables improved performance in complex scenarios.

1.1 Definition of Terms

This section provides definitions for key terms used throughout this research. By clearly defining these terms beforehand, it aims to ensure a consistent understanding of the concepts and to minimize potential confusion.

- **Imperfect information.** Imperfect information is when players in a game lack complete knowledge about the game’s state or the actions of other players (Solinas et al., 2024). Classic Imperfect-information games include most card games, where players cannot see their opponents’ cards and must make decisions based on incomplete information.
- **Perfect information.** Perfect information means all players have complete knowledge of the entire game state and the actions taken by all players at all times. In perfect-information games, there is no hidden information, and every

尽管潜力巨大，但由于多玩家环境下的不完全信息、庞大且复杂的状态与动作空间以及稀疏奖励等固有特性，为《斗地主》和《Big2》等复杂卡牌游戏训练最先进的模型仍面临挑战。本研究旨在通过分层深度强化学习（HDRL）框架提升这些游戏的性能。HDRL 是一种 DRL 方法，它将复杂任务分解为多个子任务，并采用分层结构进行学习与决策（Chai 等，2023；Gao 等，2024；Vezhnevets 等，2017）。通过将复杂任务分解为一系列更简单的子任务，每层专注于在特定层面上解决决策问题，该方法可在复杂场景中显著提升性能。

1.1 术语定义

本节为本研究中所使用的关键术语提供定义。通过预先对这些术语进行清晰界定，旨在确保对相关概念的理解保持一致，并最大限度减少潜在混淆。

- **不完全信息。**不完全信息是指游戏中的参与者对游戏状态或其他参与者的行为缺乏完整认知（Solinas等，2024）。经典的不完全信息博弈包括大多数卡牌游戏——在这些游戏中，玩家无法看到对手持有的卡牌，必须基于不完全信息做出决策。
- **完美信息。**完美信息意味着所有参与者均对整局游戏的状态以及所有参与者在各时点所采取的行动拥有完整认知。在完美信息博弈中，不存在隐

player can see the moves made by others and the current state of the game. Examples of perfect-information games include Chess, Shogi, and Go, where each player can see the entire board and all pieces at all times (Schrittwieser et al., 2020).

- **Agent.** An agent in Deep Reinforcement Learning (DRL) is the entity that interacts with the environment to learn optimal behaviors through trial and error (Milani et al., 2024; Sutton and Barto, 2018). It observes the current state, selects actions based on its policy, and receives rewards or penalties from the environment, which guide its future decisions. For example, in a self-driving car scenario, the agent is the AI system controlling the car. It processes sensor data to understand its surroundings, chooses actions like accelerating or braking, and learns from the outcomes (safe driving or collisions) to improve its driving strategy.
- **Model.** The model in DRL is the neural network that approximates the policy or value function (Milani et al., 2024; Sutton and Barto, 2018). This neural network outputs action probabilities or value estimates, which the agent uses to make decisions. For example, in Proximal Policy Optimization (PPO), the model is a neural network that takes a state (s) as input and outputs the probabilities of possible actions. In Deep Monte Carlo (DMC), the model is the neural network that estimates Q-values for state-action pairs.
- **State space.** The state space is the set of all possible configurations or states that a system can be in (Lu et al., 2024; Tierney, 1995). Each state in this space represents a unique configuration of the system's variables. In DouDiZhu,

藏信息，每位参与者均可观察其他参与者的着法及游戏的当前状态。完美信息博弈的典型示例包括国际象棋、将棋和围棋——在这些游戏中，每位参与者均可随时观察整个棋盘及所有棋子（Schrittwieser 等，2020）。

- **智能体。**在 Deep Reinforcement 学习（DRL）中，智能体是与环境交互的实体，通过试错学习最优行为（Milani 等人，2024；Sutton 与 Barto，2018）。它观察当前状态，根据其策略选择动作，并从环境中获取奖励或惩罚，这些反馈指导其未来的决策。例如，在自动驾驶场景中，智能体即为控制车辆的AI系统。它处理传感器数据以了解周围环境，选择加速或刹车等动作，并根据结果（安全驾驶或碰撞）学习，从而优化其驾驶策略。
- **模型。**DRL 中的模型是一种用于近似策略函数或价值函数的神经网络（Milani 等人，2024；Sutton 与 Barto，2018）。该神经网络输出动作概率或价值估计值，智能体据此做出决策。例如，在近端策略优化（PPO）中，模型是一种以状态（ s ）为输入、输出可能动作概率的神经网络；而在深度蒙特卡洛（DMC）中，模型则是用于估计状态-动作对的 Q 值的神经网络。
- **状态空间。**状态空间是指系统可能处于的所有配置或状态的集合（Lu 等人，2024；Tierney，1995）。该空间中的每个状态均代表系统变量的一种独特配

the state space includes all possible game states at any given time, such as the distribution of cards among the three players, current hand, sequence of played cards, player roles (landlord or peasant), and other game-specific variables like the current number of bombs. This space is highly dynamic and can be extremely large, especially in games with many cards and complex rules.

- **Action space.** The action space is the set of all possible actions that an agent can take in a given environment (Ming et al., 2023; Sutton and Barto, 2018). Each action in this space represents a distinct move or decision that an agent can make at a particular point in time. In DouDiZhu, the action space includes all possible plays that a player can make during their turn, such as playing a single card, a pair, a straight, a bomb, or passing the turn. This space is influenced by the rules of the game, the current state of play, and the cards available in the player's hand. The action space can be vast and complex, particularly in games with many potential combinations and strategies.
- **Sparse reward.** A sparse reward in DRL is when the agent receives feedback (rewards) infrequently or only under specific conditions (Wu and Chen, 2024). Unlike dense rewards, where the agent frequently receives positive or negative feedback based on its actions, sparse rewards provide limited feedback, often only when the agent achieves a significant milestone or completes the task. In DouDiZhu and Big2, a sparse reward scenario means the agent only receives a reward at the end of the game based on whether it won or lost, rather than receiving feedback after each move. This type of reward makes learning more challenging because the agent has fewer signals to guide its actions and must rely more on exploring the state space to find successful action sequences.

置。在《斗地主》中，状态空间包含任意给定时刻下所有可能的游戏状态，例如三名玩家间的卡牌分布、当前手牌、已出卡牌的序列、玩家角色（地主或农民），以及游戏特定变量（如当前炸弹数量）。该空间具有高度动态性，且规模可能极为庞大——尤其在卡牌数量众多、规则复杂的游戏中。

- **动作空间。**动作空间是指智能体在给定环境中可采取的所有可能动作的集合（Ming 等人，2023；Sutton 与 Barto，2018）。

在该空间中的每个动作，都代表智能体在特定时刻可做出的一个独立决策或操作。在 DouDiZhu 游戏中，动作空间包含玩家在其回合内可采取的所有可能操作，例如打出单张牌、一对牌、顺子、炸弹，或放弃回合。该空间受游戏规则、当前游戏状态以及玩家手中可用牌张的影响。对于包含大量潜在组合与策略的游戏而言，其动作空间可能极为庞大且复杂。

- **稀疏奖励。**在 DRL 中，稀疏奖励是指智能体很少或仅在特定条件下（Wu 和 Chen，2024）获得反馈（奖励）。

与密集奖励不同——在密集奖励场景中，智能体根据其行为频繁获得正向或负向反馈——稀疏奖励仅提供有限的反馈，且通常仅在智能体达成重要里程碑或完成任务时才会出现。在 DouDiZhu 和 Big2 中，稀疏奖励场景意味着智能体仅在游戏结束时根据胜负结果获得奖励，而非在每次移动后都获得反馈。这种奖励机制增加了学习难度，因为智能体可用于指导其行为的信号较少，必须更多地依赖对状态空间的探索来寻找成功的动作序列。

1.2 Research Motivation

The motivation for researching complex imperfect-information card games includes several key aspects. Firstly, these games provide a challenging environment to explore large state and action spaces and understand decision-making in uncertain scenarios (Bekius et al., 2022; Rumeser and Emsley, 2019; Świechowski and Tajmajer, 2021). Secondly, developing AI agents that excel in these games pushes the boundaries of AI capabilities in decision-making under uncertainty. These games serve as benchmarks for evaluating and comparing AI algorithms, driving improvements and advancing the field (Schmid et al., 2023; Solinas et al., 2023). Thirdly, the significant challenges that these games present for AI motivate the creation of new techniques to handle complexity, uncertainty, and large state-action spaces (Yang et al., 2023). Lastly, insights from studying these games can be applied to finance, cybersecurity, negotiation strategies, and military planning, demonstrating the research’s broader impact (Henderson, 2021).

1.3 Problem Statement

Deep Reinforcement Learning (DRL) is an algorithm that combines Deep Neural Networks (DNNs) and Reinforcement Learning (RL) (Oroojlooy and Hajinezhad, 2023; Sutton and Barto, 2018; Yin et al., 2024). Among the most prominent DRL algorithms are Deep Q-Network (DQN) (Mnih et al., 2015; Nikpour et al., 2024), Asynchronous Advantage Actor-Critic (A3C) (Cho et al., 2024; Mnih et al., 2016), and Proximal Policy Optimization (PPO) (Schulman et al., 2017; Son et al., 2024). These algorithms achieve superior results in small-scale (less than 20 actions) Atari games. However, DouDiZhu and Big2 have more than 10^4 actions, in comparison to Atari. Consequently, when

1.2 研究动机

研究复杂不完全信息卡牌游戏的动机包含以下几个关键方面。首先，这类游戏为探索大规模状态与动作空间、以及在不确定情境下理解决策过程提供了极具挑战性的研究环境（Bekius 等人，2022；Rumeser 与 Emsley，2019；Swiechowski 与 Tajmayer，2021）。

其次，开发在这些游戏中表现卓越的AI智能体，拓展了AI在不确定性环境下决策能力的边界。这些游戏可作为评估和比较AI算法的基准，推动算法改进并推动该领域发展（Schmid等人，2023；Solinas等人，2023）。第三，这些游戏为AI带来的重大挑战，促使研究者开发应对复杂性、不确定性及大规模状态-动作空间的新技术（Yang 等人，2023）。最后，从这些游戏中获得的洞见可应用于金融、网络安全、谈判策略及军事规划等领域，彰显了该研究更广泛的潜在影响（Henderson，2021）。

1.3 问题描述

深度强化学习（DRL）是一种将深度神经网络（DNN）与强化学习（RL）相结合的算法（Oroojlooy 与 Hajinezhad，2023；Sutton 与 Barto，2018；Yin 等人，2024）。

其中最具代表性的 DRL 算法包括深度 Q 网络（DQN）（Mnih 等人，2015；Nikpour 等人，2024）、异步优势演员-评论家算法（A3C）（Cho 等人，2024；Mnih 等人，2016）以及近端策略优化算法（PPO）（Schulman 等人，2017；Son 等人，2024）。这些算法在小规模（少于 20 个动作）的 Atari 游戏中取得了优异性能。然而，与 Atari 相比，DouDiZhu 和 Big2 的游戏包含超过 10^4 个动作。因此，当将 DQN、A3C 和

DQN, A3C, and PPO are applied to these games, they may fail to converge due to the large number of output actions. Specifically, when the PPO method is applied to Big2, it only manages to defeat amateur players (Charlesworth, 2018). Meanwhile, Yang et al. (Yang et al., 2022a) and Zha et al. (Zha et al., 2021b) find that PPO, DQN, and A3C algorithms all struggle to achieve competitive performance in DouDiZhu. You et al. (You et al., 2020) propose Combination Deep Q-Learning (CQL), a two-phase DQN algorithm that decouples actions into decomposition selection and final move selection. CQL outperforms DQN and A3C in DouDiZhu (You et al., 2020). However, another study shows that CQL suffers from overestimation bias and has no advantage over rule-based heuristics AI (Zha et al., 2021b). To eliminate overestimation bias, DouZero (Zha et al., 2021b) adopts Deep Monte Carlo (DMC), a type of DRL algorithm that combines the conventional Monte Carlo method with DNNs. In a Monte Carlo self-play framework, DNNs first estimate the value of each action (Q-value), then select the action with the highest Q-value as a final move. DouZero is the forefront AI system in DouDiZhu for its remarkable performance in contrast to prior works. However, there are three challenges in DRL in complex imperfect-information card games:

- **Learning Efficiency:** Complex card games are multiplayer games with imperfect information where players have private, hidden cards. Therefore, players must decide on actions under uncertain conditions, which are more difficult than when they have perfect information and require more iterations and time. The state-of-the-art DMC system (a type of DRL algorithm) for DouDiZhu (Zha et al., 2021b) takes four GPUs and 48 CPUs for 30 days to train the model. The resource requirements and training time pose challenges for small-scale laboratories, individual researchers, and rapid deployment. This research addresses this

PPO 应用于此类游戏时，由于输出动作数量庞大，这些算法可能无法收敛。具体而言，当将 PPO 方法应用于 Big2 时，它仅能击败业余玩家（Charlesworth, 2018）。与此同时，Yang 等人（Yang et al., 2022a）与 Zha 等人（Zha et al., 2021b）发现，PPO、DQN 及 A3C 算法在 DouDiZhu 中均难以取得具有竞争力的性能。You 等人（You et al., 2020）提出了组合深度 Q 学习算法（CQL），这是一种两阶段 DQN 算法，将动作分解为分解选择与最终决策选择。CQL 在 DouDiZhu 中优于 DQN 与 A3C（You et al., 2020）。然而，另一项研究表明，CQL 存在高估偏差，且相较于基于规则的启发式 AI 并无优势（Zha et al., 2021b）。为消除高估偏差，DouZero（Zha et al., 2021b）采用了深度蒙特卡洛算法（DMC），这是一种 DRL 算法。

将传统的蒙特卡洛方法与深度神经网络相结合。在蒙特卡洛自玩框架中，深度神经网络首先估计每个动作的价值（Q值），然后选择具有最高Q值的动作作为最终行动。DouZero 是 DouDiZhu 中领先的 AI 系统，其卓越性能与先前研究相比尤为突出。然而，在复杂的不完全信息卡牌游戏中，DRL 仍面临以下三个挑战：

- **学习效率：**复杂卡牌游戏属于多玩家博弈，且信息不完全，玩家持有私有隐藏卡牌。因此，玩家必须在不确定条件下做出决策，这比拥有完整信息时更为困难，需要更多迭代次数与时间。针对《斗地主》开发的最新 DMC 系统（一种 DRL 算法）（Zha 等，2021b）需使用 4 块 GPU 与 48 块 CPU，历时 30 天才能训练完成模型。此类资源需求与训练时长对小型实验室、独立研究人员及快速部署场景构成挑战。本研究通过提出 “Oracle 引

by proposing Oracle Guiding to improve learning efficiency, which is the first research objective (Objective 1). Oracle Guiding refers to using information that is unavailable during execution but accessible during training. In this approach, agents are initially trained with all information, gradually reducing the amount of hidden information as the training progresses.

- **Learning Stability:** Complex card games have a large and complex state and action space. With the combination of cards and the complex rules, DouDiZhu has approximately 10^{83} potential states and 27,472 possible actions, while Big2 has about 10^{28} states and 17,859 possible legal actions. This complex state and action space can lead to instability in neural network models due to increased complexity, data sparsity making some actions hard to estimate, and a higher risk of overfitting. The learning process of DMC (a type of DRL algorithm) in DouDiZhu and Big2 exhibits high and fluctuating loss (He, 2022; Wang et al., 2023a). When a learning process has high and fluctuating loss, it means the model is highly sensitive to small variations in the training data. Keeping low and stable loss ensures the training process is robust to these perturbations, allowing the agent to learn reliable policies (Hansen et al., 2021). Thus, this presents a new challenge in maintaining low and stable loss to improve training stability in DMC. This research tackles this issue by introducing Adaptive Deep Monte Carlo (ADMC) to enhance learning stability (low and stable loss), which is the second research objective (Objective 2).
- **Performance Improvement:** Complex card game environments typically have sparse rewards, with no instant rewards until the game is over. This scarcity of rewards makes performance and exploration challenging for some DRL methods.

导”策略来提升学习效率，将其设为首要研究目标（目标1）。“Oracle 引导”意指利用执行阶段不可获取、但训练阶段可获取的信息。在此方法中，智能体初始阶段使用全部信息进行训练，随后随着训练进程逐步减少隐藏信息的量。

- **学习稳定性：**复杂卡牌游戏拥有庞大且复杂的状态与动作空间。结合卡牌组合与复杂规则，斗地主拥有约 10^{83} 种潜在状态及27,472种可行动作，而Big2则拥有约 10^{28} 种状态及17,859种合法动作。这种复杂的状态与动作空间可能引发神经网络模型的不稳定性——源于复杂度增加、数据稀疏性导致部分动作难以估计，以及过拟合风险升高。DMC（一种DRL算法）在斗地主与Big2中的学习过程均表现出高波动性损失（He, 2022; Wang等, 2023a）。当学习过程出现高波动性损失时，意味着模型对训练数据中的微小变化高度敏感；而保持低且稳定的损失则能确保训练过程对这些扰动具有鲁棒性，从而使智能体学会可靠策略（Hansen等, 2021）。因此，如何在DMC中维持低且稳定的损失，成为提升训练稳定性的新挑战。本研究通过引入自适应深度蒙特卡洛方法（ADMC）来增强学习稳定性（低且稳定的损失），从而解决这一问题，这构成了第二个研究目标（目标2）。
- **性能提升：**复杂的卡牌游戏环境通常具有稀疏奖励特性，在游戏结束前无法获得即时奖励。这种奖励稀缺性使得某些DRL方法在性能与探索方面面临挑战。研究人员一直致力于提升人工智能在不完全信息卡牌游戏

Researchers have been striving to improve the performance of AI in imperfect-information card games, using methods ranging from rule-based approaches to tree search algorithms and DRL techniques (Brown and Sandholm, 2019; He, 2022). Despite breakthroughs in other games like Go, Texas Hold'em, and StarCraft II (Souchleris et al., 2023), DRL still faces challenges in handling the large state-action spaces and sparse rewards of complex card games like DouDiZhu and Big2. Methods like DouZero+ (Zhao et al., 2023b) and WagerWin (Wang et al., 2023a), based on DouZero, show only small improvements. This research first improves the performance by incorporating Relative Advantage Reward Shaping (RARS) to generate reward signals for the Generalized Advantage Estimator (GAE) (Jin et al., 2024; Schulman et al., 2015b) in PPO (a type of DRL algorithm), facilitating effective training in sparse reward environments (Objective 3). RARS is a key component of the Hierarchical Deep Reinforcement Learning (HDRL) framework, which also integrates Oracle Guiding and ADMC to address the performance challenge.

By addressing these challenges, the primary objective of this research is to enhance the performance for complex imperfect-information card games through the integration of Oracle Guiding, ADMC, and RARS into a HDRL framework.

1.4 Research Questions

The research questions of this research can be summarized as follows:

- How to improve learning efficiency (reducing the training time) in Deep Monte

中的性能，所采用的方法涵盖基于规则的策略、树搜索算法以及 DRL 技术（Brown 与 Sandholm, 2019; He, 2022）。尽管在围棋、德州扑克及《星际争霸II》等游戏中已取得突破（Souchleris 等, 2023），DRL 在处理《斗智》与《Big2》等复杂卡牌游戏的大规模状态-动作空间及稀疏奖励方面仍面临挑战。基于 DouZero 开发的 DouZero+（Zhao 等, 2023b）与 WagerWin（Wang 等, 2023a）等方法仅展现出有限提升。本研究首先通过引入相对优势奖励塑造（RARS）技术，为 PPO（一种 DRL 算法）中的广义优势估计器（GAE）（Jin 等, 2024; Schulman 等, 2015b）生成奖励信号，从而提升性能，助力在稀疏奖励环境中实现有效训练（目标3）。RARS 是分层深度强化学习（HDRL）框架中的关键组件，该框架还整合了 Oracle Guiding 与 ADMC，以应对性能挑战。

通过应对这些挑战，本研究的主要目标是通过将 Oracle Guiding、ADMC 和 RARS 整合到 HDRL 框架中，提升复杂不完全信息卡牌游戏的性能。

1.4 研究问题

本研究的研究问题可概括如下：

- 如何提升 Deep Monte 的学习效率（缩短训练时间）

Carlo (DMC) training?

- How to improve the learning stability (keeping low and stable loss) in Deep Monte Carlo (DMC) training?
- How to improve the performance (winning score) for complex card games with imperfect information?

1.5 Research Objectives

The primary objective of this research is to enhance the performance for complex imperfect-information card games by integrating proposed methods into a Hierarchical Deep Reinforcement Learning (HDRL) framework. To achieve this main goal, the research is broken down into the following sub-research objectives:

- To improve learning efficiency in Deep Monte Carlo (DMC) training by introducing Oracle Guiding, which accelerates the training process. In Oracle Guiding, The Oracle agent initially trains with full information and gradually transitions to a standard agent by incrementally discarding imperfect information (discussed in Section 3.3 in Chapter 3).
- To improve learning stability in Deep Monte Carlo (DMC) training by proposing the Adaptive Deep Monte Carlo Method (ADMC), which incorporates Q-value weight clipping to prevent drastic policy changes and reduce high variance during training (discussed in Section 3.4 in Chapter 3).
- To improve the performance of Proximal Policy Optimization (PPO) in complex imperfect-information card games by designing Relative Advantage Reward

Carlo (DMC) 培训?

- 如何在深度蒙特卡洛 (DMC) 训练中提升学习稳定性 (保持较低且稳定的损失值)?
- 在具有不完全信息的复杂卡牌游戏中, 如何提升游戏性能 (获胜分数)?

1.5 研究目标

本研究的主要目标是通过将提出的方法整合到分层深度强化学习 (HDRL) 框架中, 提升复杂不完全信息卡牌游戏的性能。为实现这一主要目标, 本研究将其分解为以下子研究目标:

- 通过引入可加速训练过程的 “Oracle Guiding” 机制, 提升深度蒙特卡洛 (DMC) 训练中的学习效率。在 Oracle Guiding 中, Oracle 智能体最初基于完整信息进行训练, 随后逐步通过渐进式剔除不完美的信息, 过渡至标准智能体训练模式 (详见第3.3章第3节)。
- 为了提升深度蒙特卡洛 (DMC) 训练中的学习稳定性, 本文提出了自适应深度蒙特卡洛方法 (ADMC), 该方法结合了Q值权重裁剪技术, 以防止策略的剧烈变化, 并降低训练过程中的高方差 (详见第3.4节, 第3章)。
- 为了提升近端策略优化 (PPO) 在复杂不完全信息卡牌游戏中的性能, 本文设计了相对优势奖励塑造 (RARS) 方法, 该方法为 PPO 生成即时奖励, 从而使其在稀疏奖励环

Shaping (RARS), which generates instant rewards for PPO to effectively learn in sparse reward environments (discussed in Section 3.5 in Chapter 3).

Integrating Oracle Guiding, ADMC, and RARS into the HDRL framework aims to further enhance the performance in complex imperfect-information card games, thereby achieving the primary objective of this research (discussed in Section 3.6 in Chapter 3).

1.6 Research Scope

This study focuses on complex imperfect-information card games, using DouDiZhu and Big2 as case studies detailed in B.1 and B.2. DouDiZhu and Big2 are valuable not only because these games are two typical representatives of imperfect-information card games but also because they are extremely popular in China and Southeast Asia, attracting millions of players daily. They have three distinctive features that attract researchers. First, DouDiZhu and Big2 are imperfect-information games with multiple players. Players can only see their own cards and the public cards; they cannot see their opponents' hands. Second, DouDiZhu and Big2 have a large and complex state and action space. Due to the combination of cards and the complex rules, DouDiZhu has approximately 10^{83} potential states and 27,472 possible actions, while Big2 has about 10^{28} states and 17,859 possible legal actions. Third, DouDiZhu and Big2 are games with sparse rewards. There are no instant rewards in each round of the confrontation between different camps until the game is over.

By exploring the specific research scopes of DouDiZhu and Big2, this study ad-

境中有效学习（详见第3.5节，第3章）。

将 Oracle Guiding、ADMC 和 RARS 整合到 HDRL 框架中，旨在进一步提升复杂不完全信息卡牌游戏的性能，从而实现本研究的核心目标（详见第3.6节，第3章）。

1.6 研究范围

本研究聚焦于复杂不完全信息卡牌游戏，并以《DouDiZhu》与《Big2》作为案例研究，详见B.1与B.2部分。这两款游戏之所以具有重要研究价值，不仅在于它们是不完全信息卡牌游戏的典型代表，更在于它们在中国及东南亚地区广受欢迎，每日吸引数百万玩家参与。它们具备三项吸引研究者的显著特征：首先，DouDiZhu与Big2均为多玩家不完全信息博弈，玩家仅能查看自身手牌与公共牌，而无法窥见对手的手牌；其次，DouDiZhu与Big2均具有庞大且复杂的状态空间与动作空间。

由于卡牌组合及复杂规则的影响，DouDiZhu 拥有约 1083 种潜在状态和 27,472 种可行行动，而 Big2 则拥有约 1028 种状态及 17,859 种合法可行行动。第三，DouDiZhu 与 Big2 均属于稀疏奖励类游戏——在双方阵营对战的每轮对局中，直至游戏结束，均不存在即时奖励。

本研究通过探究 DouDiZhu 与 Big2 的具体研究范畴，深化了对这两类复杂不完全信

vances the understanding of these complex imperfect-information card games. It develops effective algorithms and creates tailored AI agents for each game. Furthermore, it evaluates performance and analyzes agent behavior in both games.

1.7 Research Contribution

There are three research contributions in this research. First, this research proposes Oracle Guiding to improve the learning efficiency of Deep Monte Carlo (DMC) for complex imperfect-information card games. In Oracle Guiding, the Oracle agent is initially trained using both imperfect information (the private cards of other players) and perfect information (its own cards and public cards). During training, the Oracle agent transitions to using only its own cards and public cards, unlike a standard agent using only perfect information. Ablation experiments demonstrate that Oracle Guiding learns faster than the original method on the same server within the same training time.

The second research contribution is the proposal of the Adaptive Deep Monte Carlo Method (ADMC). To mitigate high and fluctuating loss during DMC training, gradient weight clipping is introduced in ADMC to ensure moderate policy updates and prevent drastic changes. Ablation experiments demonstrate that ADMC achieves better stability than the original method. The combination of Oracle Guiding and ADMC, named OADMCDou for DouDiZhu and OADMCDBig for Big2, shows improved performance and serves as a sub-value model in a Hierarchical Deep Reinforcement Learning (HDRL) framework.

Lastly, the third research contribution is the improvement in performance for complex imperfect-information card games through the integration of Relative Advantage

息卡牌游戏的理解；开发了高效算法，并为每种游戏构建了定制化人工智能智能体；

此外，本研究还评估了这两种游戏中的性能表现，并分析了智能体的行为特征。

1.7 研究贡献

本研究包含三项学术贡献。首先，本研究提出“Oracle Guiding”方法，旨在提升深度蒙特卡洛算法（DMC）在复杂不完全信息卡牌游戏中的学习效率。在

“Oracle Guiding”框架中，Oracle 智能体最初是结合不完全信息（其他玩家的私有卡牌）进行训练的。

以及完整信息（包括自身持有的牌与公开牌）。在训练过程中，Oracle 智能体转而仅使用自身持有的牌与公开牌，这与仅使用完整信息的标准智能体不同。消融实验表明，在相同服务器及相同训练时间内，Oracle Guiding 的学习速度优于原始方法。

第二项研究贡献是提出了自适应深度蒙特卡洛方法（ADMC）。为了缓解 DMC 训练过程中出现的高且波动的损失，ADMC 引入了梯度权重裁剪，以确保策略更新的适度性，防止剧烈变化。消融实验表明，ADMC 相比原始方法具有更好的稳定性。将“Oracle 引导”与 ADCMC 相结合，分别命名为 OADMCDou（用于 DouDiZhu）和 OADMMCBig（用于 Big2），这些方法不仅提升了性能，还在分层深度强化学习（HDRL）框架中充当子价值模型。

最后，第三项研究贡献是通过将相对优势奖励塑造（RARS）、神谕引导和 ADCMC 整合到 HDRL 框架中，提升了复杂不完全信息卡牌游戏的性能。在该框

Reward Shaping (RARS), Oracle Guiding, and ADMC into the HDRL framework. In this framework, the main-policy model, trained with PPO and RARS, makes high-level strategic decisions that are further refined into specific actions by the sub-value model, trained with Oracle Guiding and ADMC. Experiments comparing this framework with state-of-the-art benchmarks for DouDiZhu and Big2 highlight its significant performance improvements.

1.8 Thesis Organization

The thesis is structured into five chapters, each building on the previous to enhance the performance in complex imperfect-information card games through a Hierarchical Deep Reinforcement Learning (HDRL) framework.

Chapter 1: Introduction. Provides an overview of the significance and challenges of applying Artificial Intelligence (AI) to complex imperfect-information card games. It explains the breakthroughs of Deep Reinforcement Learning (DRL) in game playing, especially in perfect information games, and highlights the difficulties DRL faces in imperfect information card games. This chapter also defines key terms, outlines the research motivation, problem statement, research questions, and objectives, and delineates the research scope and contributions.

Chapter 2: Literature Review. Reviews the existing work related to advancements and challenges in DRL applied to complex imperfect-information card games like DouDiZhu and Big2. It covers key DRL concepts, including Monte Carlo methods, Temporal-Difference methods, Actor-Critic methods, Proximal Policy Optimization, and Hierarchical Deep Reinforcement Learning. This chap-

架中，主策略模型使用PPO和 RARS 训练，做出高层次的战略决策；这些决策随后由次级价值模型——使用神谕引导和 ADMC 训练——细化为具体行动。将这一框架与DouDiZhu和Big2领域的最新基准模型进行对比实验的结果表明，该框架在性能上取得了显著提升。

1.8 论文结构

本论文共分为五个章节，每章均在前一章的基础上展开，旨在通过分层深度强化学习（HDRL）框架，提升复杂不完全信息卡牌游戏的性能。

第1章：引言。本章概述了将人工智能（AI）应用于复杂不完全信息卡牌游戏的重要性及面临的挑战。文中阐述了深度强化学习（DRL）在卡牌游戏领域，尤其是完全信息卡牌游戏中的突破性进展，并强调了 DRL 在不完全信息卡牌游戏中面临的困难。本章还定义了关键术语，概述了研究动机、问题陈述、研究问题与目标，以及研究范围与贡献。

第2章：文献综述。本章回顾了与 DRL 技术在复杂不完全信息卡牌游戏（如《斗地主》和《Big2》）中的应用进展及挑战相关的现有研究。涵盖了 DRL 技术的关键概念，包括蒙特卡洛方法、时间差分方法、行动者-评论家方法、近端策略优化以及分层深度强化学习。本章还综述了 DRL 技

ter also reviews representative applications of DRL in card games, discusses techniques for enhancing efficiency, stability, and performance, and identifies current research gaps, suggesting new approaches like Oracle Guiding, Adaptive Deep Monte Carlo (ADMC), and Relative Advantage Reward Shaping (RARS).

Chapter 3: Method Design and Implementation. Explains the design and implementation of the Oracle Guiding, ADMC, and RARS methods. It addresses the research questions and achieves objectives from Chapter 1. Furthermore, Chapter 3 details how these methods are integrated into the HDRL framework to improve the performance for complex imperfect-information card games.

Chapter 4: Experiment Results and Discussion. Evaluates the methods discussed in Chapter 3, including Oracle Guiding, ADMC, RARS, and the HDRL framework. It demonstrates the advantages of these methods in terms of learning efficiency, stability, and performance through experiment validation. Additionally, case studies illustrate the practical benefits of the HDRL framework in DouDiZhu and Big2 card games. Chapter 4 confirms the achievement of the research objectives and addressing the research questions outlined in Chapter 1.

Chapter 5: Conclusion and Future Work. Summarizes the main contributions and findings, discusses limitations, and suggests future research directions to further advance the performance in complex imperfect-information card games.

术在卡牌游戏中的代表性应用，探讨了提升效率、稳定性和性能的 techniques，并指出了当前研究中的空白，提出了诸如Oracle Guiding、自适应深度蒙特卡洛（ADMC）以及相对优势奖励塑造（RARS）等新方法。

第3章：方法设计与实现。本章阐述了Oracle Guiding、ADMC及RARS方法的设计与实现，回应了研究问题，并实现了第1章设定的目标。此外，第3章详细说明了这些方法如何整合到 HDRL 框架中，以提升复杂不完全信息卡牌游戏的性能。

第4章：实验结果与讨论。本章评估了第3章中讨论的方法，包括Oracle Guiding、ADMC、RARS以及HDRL框架。通过实验验证，本章展示了这些方法在学习效率、稳定性和性能方面的优势。此外，案例研究还阐明了HDRL框架在DouDiZhu和Big2卡牌游戏中的实际应用价值。第4章确认了研究目标的实现，并回答了第1章中提出的研究问题。

第5章：结论与未来研究方向。本章总结了主要贡献与研究结果，探讨了研究局限性，并提出了未来研究方向，以进一步推动复杂不完全信息卡牌博弈的性能提升。

CHAPTER 2

LITERATURE REVIEW

2.1 Introduction

Recent advancements in game Artificial Intelligence (AI) have led to the development of sophisticated systems, transitioning from rudimentary agents such as the DQN Atari (Mnih et al., 2015) to more advanced systems including AlphaZero (Silver et al., 2017), Libratus (Brown and Sandholm, 2018), OpenAI Five (Berner et al., 2019), and AlphaStar (Vinyals et al., 2019). These AI systems surpass human experts, indicating a profound enhancement in decision-making. For instance, AlphaZero outperforms professional Go players in perfect-information games (PIGs), while OpenAI Five excels in imperfect-information games (IIGs), becoming the first AI to defeat world champions in an eSports competition. PIGs are games in which players can observe all game states; in contrast, IIGs refer to games in which participants cannot see the complete information of other players. With AlphaStar and OpenAI Five defeating professional players in StarCraft II (Vinyals et al., 2019) and Dota2 (Berner et al., 2019), it is evident that current AI techniques master complex games. This proficiency is not limited to these examples. Recent AI advancements in games like Honor of Kings (Yang et al., 2022b) use methods similar to AlphaStar and OpenAI Five, indicating a shared foundation in AI approaches across various game genres. These genres span board games like Go (Schrittwieser et al., 2020), Stratego (Perolat et al., 2022), and Tic-Tac-Toe (van Opheusden et al., 2023); card games such as Heads-up No-limit Texas Hold'em (HUNL) (Wang et al., 2023b), DouDiZhu (Jiang et al., 2019), and Mahjong (Wang

第 2 章 文献综述

2.1 引言

游戏人工智能 (AI) 领域的最新进展催生了诸多精密系统的开发, 从诸如 DQN Atari (Mnih 等人, 2015) 这类基础智能体, 逐步发展到包括AlphaZero (Silver等人, 2017)、Libratus (Brown 与 Sandholm, 2018)、OpenAI Five (Berner等人, 2019) 以及AlphaStar (Vinyals等人, 2019) 等更先进的系统。这些AI系统已超越人类专家, 表明其在决策能力上取得了质的飞跃。例如, AlphaZero在完美信息游戏 (PIGs) 中表现优于职业围棋选手; 而OpenAI Five则在非完美信息游戏 (IIGs) 中表现出色, 成为首个在电子竞技赛事中击败世界冠军的AI。PIGs是指玩家可观察所有游戏状态的游戏; 相比之下, IIGs则指参与者无法获知其他玩家完整信息的游戏。AlphaStar与OpenAI Five已在《星际争霸II》(Vinyals等人, 2019) 及《Dota2》(Berner 等人 (2019) 的研究表明, 当前的人工智能技术已能驾驭复杂游戏。这种能力并不局限于上述案例。近期在《王者荣耀》等游戏中取得的AI进展 (Yang 等人, 2022b) 采用了与 AlphaStar 及 OpenAI Five 相似的方法, 这表明不同游戏类型中的AI技术具有共同的理论基础。这些游戏类型包括围棋等棋盘游戏 (Schrittwieser 等人, 2020)、Stratego (Perolat 等人, 2022) 与井字棋 (van Opheusden 等人, 2023); 纸牌游戏如 Heads-up No-limit Texas Hold 'em (HUNL) (Wang 等人, 2023b)、斗地主 (Jiang 等人, 2019) 与麻将 (Wang等人, 2022

et al., 2022a); first-person shooters represented by Quake III Arena (Jaderberg et al., 2019); and real-time strategy games including StarCraft II, Dota2, and Honor of Kings. The AI behind these achievements include AlphaGo (Silver et al., 2016), DeepStack (Moravčík et al., 2017), AlphaGo Zero (Silver et al., 2017), AlphaZero (Silver et al., 2018), FTW (Jaderberg et al., 2019), Suphx (Li et al., 2020), JueWu (Ye et al., 2020), Commander (Wang et al., 2021), DouZero (Zha et al., 2021b), DeepNash (Perolat et al., 2022), WagerWin (Wang et al., 2023a), Cognitive Model (van Opheusden et al., 2023) and DanZero+ (Zhao et al., 2024b). Given these advancements, especially in card games, it naturally raises the question: What are the possible challenges of current techniques in complex imperfect-information card games? What is the choice of research methodology, what are the gaps, and what is the motivation for this study? This chapter aims to review recent successful AIs in complex imperfect-information card game and tries to answer these question through a thorough analysis of current techniques.

The remainder of this chapter unfolds as follows: Section 2.2 introduces the underlying theories related to Deep Reinforcement Learning (DRL). Section 2.3 reviews representative applications of complex imperfect-information card games, focusing on previous studies about AI algorithms in DouDiZhu and Big2. Following that, Section 2.4 discusses techniques for enhancing efficiency, stability, and performance in DRL. Finally, Section 2.5 identifies research gaps and motivates the study by summarizing the advantages and limitations of existing methods and suggesting new methods to improve them.

a) ; 以《Quake III Arena》为代表的第一人称射击游戏 (Jaderberg 等人, 2019) ; 以及包括《星际争霸II》、《Dota2》与《王者荣耀》在内的即时战略游戏。这些成就背后的AI技术包括 AlphaGo (Silver 等人, 2016) 、DeepStack (Moravcik 等人, 2017) 、AlphaGo Zero (Silver 等人, 2017) 、AlphaZero (Silver 等人, 2018) 、FTW (Jaderberg 等人, 2019) 、Suphx (Li 等人, 2020) 、JueWu (Ye 等人, 2020) 、Commander (Wang 等人, 2021) 、DouZero (Zha 等人, 2021b) 、DeepNash (Perolat 等人, 2022) 、WagerWin (Wang 等人, 2023a) 、Cognitive Model (van Opheusden 等人, 2023) 以及 DanZero+ (Zhao 等人, 2024b) 。鉴于这些进展, 尤其是在卡牌游戏中, 自然引出了以下问题: 当前技术在复杂不完全信息卡牌游戏中的潜在挑战是什么? 应选择何种研究方法? 存在哪些研究空白? 本研究的动机又是什么? 这一问pter 旨在回顾复杂不完全信息卡牌游戏中近期取得成功的人工智能应用案例, 并通过对现有技术的全面分析, 试图解答以下问题。

本章其余部分安排如下: 第2.2节介绍与深度强化学习 (DRL) 相关的基础理论; 第2.3节回顾复杂不完全信息卡牌游戏的代表性应用, 重点关注此前关于AI算法在《斗地主》和《Big2》中的研究; 随后, 第2.4节探讨提升 DRL 效率、稳定性和性能的技术; 最后, 第2.5节总结现有方法的优势与局限性, 指出研究空白, 并提出改进这些方法的新型方案, 从而为本研究提供动机。

2.2 Preliminaries

2.2.1 Deep Reinforcement Learning

Deep Reinforcement Learning (DRL) combines Deep Neural Networks (DNNs) (Krizhevsky et al., 2012; Zhao et al., 2024a) with Reinforcement Learning (RL) (Plaata et al., 2023; Sutton and Barto, 2018) to enable agents to learn complex behaviors from high-dimensional inputs. It provides the fundamental theories and methods to solve complex problems like imperfect-information card games.

2.2.1(a) Deep Neural Networks

Deep Neural Networks (DNNs) are computational models with multiple layers, inspired by the human brain's structure and function (LeCun et al., 2015). They learn complex patterns from large datasets by iteratively adjusting internal parameters. Due to the complexity and variety of states, using a simple table or other forms to record states not only consumes substantial hardware resources but also results in inefficient performance. Thus, researchers usually utilize DNNs to store the value of a state. States represent the public cards and hidden cards in imperfect information card games. In these games, Deep Reinforcement Learning (DRL) uses DNNs to approximate value functions or policies, allowing agents to learn and make decisions in complex environments by mapping states to actions or value estimates. DNNs exhibit non-linearity and scalability, making them suitable for many applications. For instance, SL (Zha et al., 2021b) for DouDiZhu and (Charlesworth, 2018) for Big2 use DNNs to approximate the value of actions in different game states.

2.2 预备知识

2.2.1 深度强化学习

深度强化学习（DRL）结合了深度神经网络（DNNs）。(Krizhevsky 等人, 2012; Zhao 等人, 2024a) 结合强化学习（RL）（Plaat 等人, 2023; Sutton 与 Barto, 2018），使智能体能够从高维输入中学习复杂行为。该方法为解决不完全信息卡牌游戏等复杂问题提供了基础理论与方法。

2.2.1(a) 深度神经网络

深度神经网络（DNNs）是一种具有多层结构的计算模型，其设计灵感源自人类大脑的结构与功能（LeCun 等人, 2015）。它们通过迭代调整内部参数，从大规模数据集中学习复杂模式。鉴于状态的复杂性与多样性，采用简单表格或其他形式来记录状态不仅会消耗大量硬件资源，还会导致性能低下。因此，研究人员通常采用 DNNs 来存储状态值。

在不完全信息卡牌游戏中，牌面与暗牌分别代表公共牌与隐藏牌。

在这些游戏中，深度强化学习（DRL）利用深度神经网络（DNN）来近似价值函数或策略，使智能体能够通过将状态映射到动作或价值估计，在复杂环境中学习和做出决策。深度神经网络具有非线性特性和可扩展性，使其适用于多种应用场景。例如，SL (Zha 等人, 2021b) 针对 DouDiZhu，以及 (Charlesworth, 2018) 针对 Big2，均使用深度神经网络来近似不同游戏状态中动作的价值。

2.2.1(b) Reinforcement Learning

Reinforcement Learning (RL) (Plaatt et al., 2023; Sutton and Barto, 2018), based on Markov Decision Process (MDP), is a mathematical framework that enables agents to learn optimal decision-making strategies by interacting with dynamic environments, as shown in Figure 2.1. In RL, an agent is an entity that learns to interact with an environment to achieve some objective. Agents in Deep Reinforcement Learning (DRL) often employ Deep Neural Networks (DNNs) to approximate complex functions that map observations to actions and value estimates, as shown in Figure 2.1. In a discounted episodic MDP denoted as (S, A, γ, P, r) , the agent chooses an action a_t according to the policy $\pi(a_t|s_t)$ at state s_t . The environment receives the action, produces a reward r_{t+1} , and transitions to the next state s_{t+1} according to the transition probability $P(s_{t+1}|s_t, a_t)$. In RL, agents aim to find a policy π that maximizes the expected cumulative reward, often denoted as the objective function $J(\pi)$. RL achieves this objective by iteratively updating the value function $V(s)$ or the action-value function $Q(s, a)$ using methods such as the Monte Carlo method (Rubinstein and Kroese, 2016; Shonkwiler and Mendivil, 2024), Temporal-Difference method (Ren et al., 2024; Sutton and Barto, 2018), Actor-Critic (Konda and Tsitsiklis, 1999; Padhye and Lakshmanan, 2023), and Proximal Policy Optimization (PPO) (Schulman et al., 2017; Son et al., 2024). Traditionally, Monte Carlo method, Temporal-Difference method, Actor-Critic, and PPO often use a table memory structure (tabular method) to store the value function of each state or each state-action pair. This approach is inefficient for complex problems with a large number of states due to significant memory constraints. The tabular method requires maintaining a table with an entry for each possible state or state-action pair, which becomes impractical as the state space grows. However, when these methods are

2.2.1(b) 强化学习

强化学习 (RL) (Plaatt等人, 2023; Sutton与Barto, 2018) 基于马尔可夫决策过程 (MDP), 是一种数学框架, 使智能体能够通过与动态环境交互来学习最优决策策略, 如图2.1所示。在强化学习中, 智能体是一个通过学习与环境交互以实现特定目标的实体。深度强化学习 (DRL) 中的智能体常采用深度神经网络 (DNN) 来近似复杂函数, 这些函数将观测值映射到动作及价值估计, 如图2.1所示。在记为 (s, A, γ, P, r) 的折扣式分阶段 MDP 中, 智能体根据策略 $\pi(a_t|s_t)$ 在状态 s_t 下选择一个动作 a_t 。环境接收动作, 生成奖励 r_{t+1} , 并根据转移概率 $P(s_{t+1}|s_t, a_t)$ 转移到下一状态 s_{t+1} 。在强化学习中, 智能体旨在寻找一个能最大化期望累积奖励的策略 π , 该策略通常记为目标函数 $J(\pi)$ 。强化学习通过迭代更新价值函数 $V(s)$ 或动作价值函数 $Q(s, a)$ 来实现这一目标, 所用方法包括蒙特卡洛方法 (Rubinstein and Kroese, 2016; Shonkwiler and Mendivil, 2024)、时序差分法 (Ren et al., 2024; Sutton and Barto, 2018)、演员-评论器算法 (Konda and Tsitsiklis, 1999; Padhye and Lakshmanan, 2023) 以及近端策略优化算法 (PPO) (Schulman 等人, 2017; Son 等人, 2024)。传统上, 蒙特卡洛方法、时序差分法、Actor-Critic 算法及 PPO 算法常采用表格记忆结构 (表格法) 来存储每个状态或每个状态-动作对的价值函数。对于具有大量状态的复杂问题, 由于内存限制严重, 这种方法效率较低。表格法需要为每个可能状态或状态-动作对维护一个条目表, 但随着状态空间扩大, 这种做法已不

combined with DNNs, they can effectively address complex problems with huge states by using the neural networks' ability to generalize and approximate functions. The Monte Carlo method and Temporal-Difference method are fundamental approaches for estimating values in RL, with Monte Carlo methods relying on complete episodes and Temporal-Difference methods updating estimates incrementally. Actor-Critic combines value-based and policy-based methods, with the critic providing value estimates and the actor adjusting the policy. PPO based on the Actor-Critic method is an optimization algorithm that constrains policy updates to prevent large policy changes. Together, they form a series of techniques used in RL for value estimation, policy learning, and optimization.

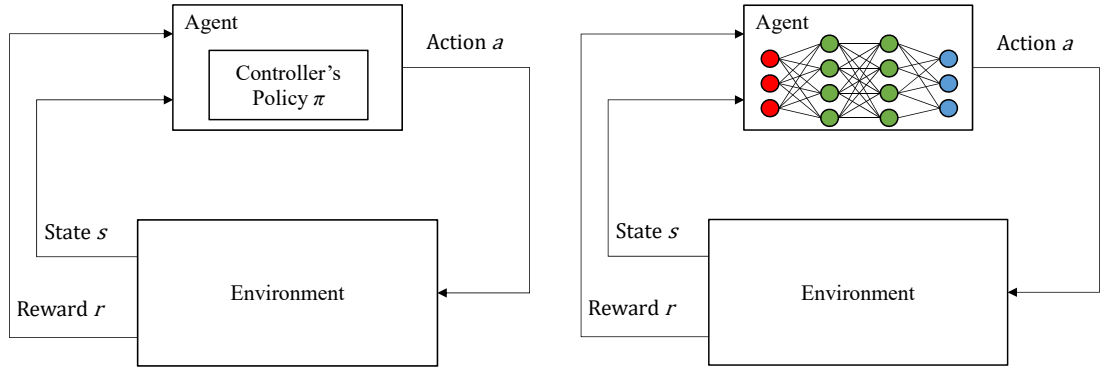


Figure 2.1: Comparison of Reinforcement Learning (left) and Deep Reinforcement Learning (DRL) (right). Agents in DRL often employ Deep Neural Networks to approximate complex functions that map observations to actions and value estimates.

Monte Carlo method. The Monte Carlo (MC) method estimates the value by repeatedly generating episodes and recording the average return at each state or each state-action pair (Binder, 2022; Luengo et al., 2020). The state value represents the expected cumulative reward starting from a specific state s_t under a given policy π , calculated as the average return from episodes starting at s_t . Similarly, the state-action

再可行。然而，若将这些方法与深度神经网络结合，则可利用神经网络泛化及函数逼近的能力，有效应对具有海量状态的复杂问题。蒙特卡洛方法与时序差分法是强化学习中原始的价值估计方法：蒙特卡洛法依赖完整回合数据，而时序差分法则通过增量方式更新估计值。Actor-Critic 算法融合了基于价值与基于策略的方法，其中“评判器”负责提供价值估计，“执行器”则负责调整策略。基于 Actor-Critic 方法的 PPO 是一种优化算法，它通过约束策略更新来防止策略发生剧烈变化。二者共同构成了强化学习领域中用于价值估计、策略学习与优化的系列技术。

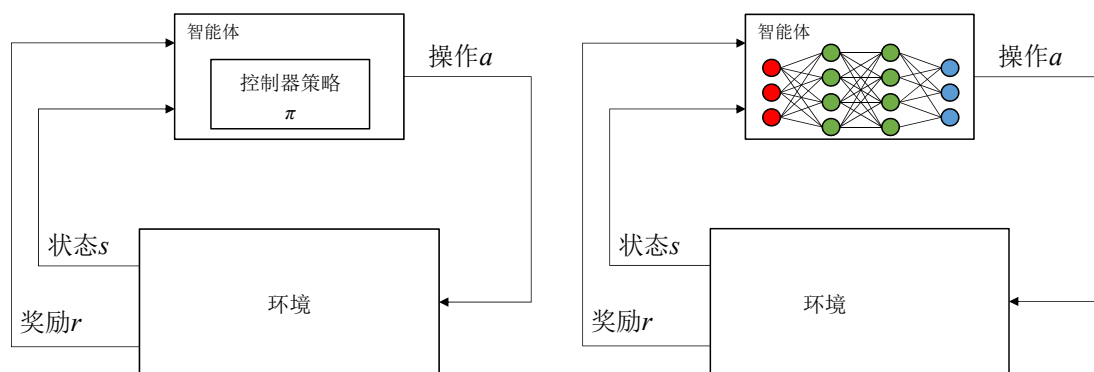


图 2.1: 强化学习（左）与深度强化学习（DRL）（右）的对比。DRL 中的智能体通常采用深度神经网络来近似复杂函数，这些函数将观测值映射到动作及价值估计。

蒙特卡洛方法。蒙特卡洛（MC）方法通过反复生成轨迹，并记录每个状态或每个状态-动作对下的平均回报，来估计价值函数（Binder, 2022; Luengo等, 2020）。状态价值表示在给定策略 π 下，从特定状态 s_t 出发的期望累积奖励，其计算方式为从 s_t 出发的所有轨迹的平均回报。类似地，状态-动作价值表示对于

value represents the expected cumulative reward for a given state-action pair (s, a) , calculated as the average return from episodes in which the agent starts at s_t and takes action a_t , as shown in Equation 2.1:

$$Q_{\pi}(s, a) = \lim_{k \rightarrow \infty} \mathbb{E}[r^k(s_t, a_t) | s_t = s, a_t = a, \pi] \quad (2.1)$$

Here, $r^k(s_t, a_t)$ denotes the observed return in episode k , and the expectation $\mathbb{E}$ is calculated by averaging the observed reward $r^k(s_t, a_t)$ from many episodes. The MC method does not require any knowledge of transition probabilities and is particularly suitable for episodic tasks like complex card games (Zha et al., 2021b). However, this method makes two assumptions to ensure convergence: a large number of episodes and frequent visits to every state and action. In this research, the Monte Carlo method serves as the foundation for the Deep Monte Carlo (DMC) method, which is enhanced using Oracle Guiding and ADMC to address the challenges of learning efficiency and stability in complex imperfect-information card games.

Temporal-difference method. Unlike the MC method, the Temporal-Difference (TD) method (Ren et al., 2024; Sutton and Barto, 2018) makes updates at every step within an episode rather than waiting until the end. This allows TD methods to update value estimates more frequently and potentially learn faster from fewer episodes. In TD methods, the state value or state-action value is updated at each step by considering the received reward and the estimated value of the next state or state-action pair, adjusting the current estimate based on the difference between expected and actual outcomes. The key difference between MC and TD methods lies in when they update their value

给定状态-动作对 (s, a) 的期望累积奖励，其计算方式为从智能体从 s_t 出发并采取动作 a 的所有轨迹的平均回报，如公式2.1所示：

$$Q \pi(s, a) = \lim_{k \rightarrow \infty} E[r_k(st, at) | s_t = s, at = a, \pi] \quad (2.1)$$

此处， $r_k(st, at)$ 表示第 k 个回合中观测到的回报，而期望值 E 则是通过对多个回合中观测到的回报 $r_k(st, at)$ 取平均值计算得出。The MC 方法无需任何关于转移概率的知识，特别适用于复杂卡牌游戏等回合制任务（Zha 等人，2021b）。然而，该方法为确保收敛性需满足两个假设：足够多的回合数，以及频繁访问每个状态与动作。本研究中，蒙特卡洛方法构成了深度蒙特卡洛（DMC）方法的基础，该方法通过引入“Oracle Guiding”与 ADMC 进行增强，以应对复杂不完全信息卡牌游戏中学习效率与稳定性方面的挑战。

时序差分法。与蒙特卡洛方法不同，时序差分法（TD）（Ren等人，2024；Sutton与Barto，2018）在每轮训练中的每一步都进行参数更新，而非等到训练结束时才更新。这使得TD方法能够更频繁地更新价值估计值，并有望用更少的训练回合更快完成学习。在TD方法中，每一步都会根据已获得的奖励以及下一状态或状态-动作对的估计值，通过比较预期结果与实际结果之间的差异，来更新当前状态值或状态-动作值估计。

MC 方法与 TD 方法的关键区别在于它们更新价值估计的时间点：MC 方法在每个回合结束

estimates. MC methods update values at the end of an episode based on total return, while TD methods update values at each time step based on expected future returns. The TD method forms the foundation of the Actor-Critic method as it uses TD error to update the Critic’s value estimate, guiding the Actor’s policy updates.

Actor-Critic. Actor-Critic consists of value function approximation (Critic) and policy optimization (Actor), where the Actor is responsible for selecting actions and the Critic evaluates the actions taken by the Actor (Konda and Tsitsiklis, 1999; Padhye and Lakshmanan, 2023). The Critic’s role is to evaluate the actions taken by the Actor by estimating the value function. To do this, the Critic uses the Temporal-Difference (TD) method, which updates the value estimates based on the TD error (δ_t). The TD error is defined as follows:

$$\delta_t = r_{t+1} + \gamma V(s_{t+1}) - V(s_t) \quad (2.2)$$

The Actor’s role is to select actions based on a policy that is parameterized by θ . The Actor uses the feedback from the Critic, specifically the TD error δ_t , to update its policy parameters. The TD error δ_t represents the temporal difference in value estimates and provides a signal for how much better or worse the action a_t is compared to the expected value (Schulman et al., 2015b; Sutton and Barto, 2018). The Actor-Critic method forms the basis for Proximal Policy Optimization (PPO), which further refines policy updates to ensure stability and improve performance.

Proximal Policy Optimization. Proximal Policy Optimization (PPO) optimizes the Actor-Critic method by introducing a way to limit the step size of policy updates, which

时基于总回报更新价值估计，而 TD 方法则在每个时间步基于预期未来回报更新价值估计。

TD 方法构成了 Actor-Critic 方法的基础，因为它利用 TD 误差来更新 Critic 的价值估计，进而引导 Actor 的策略更新。

演员-评论家算法。演员-评论家算法由价值函数近似（评论家）和策略优化（演员）组成，其中演员负责选择动作，评论家则对演员所选动作进行评估（Konda 与 Tsitsiklis, 1999; Padhye 与 Lakshmanan, 2023）。评论家的职责是通过估计价值函数来评估演员所选动作。为此，评论家采用时序差分（TD）方法，该方法基于 TD 误差（ δ_t ）更新价值估计值。TD 误差定义如下：

$$\delta_t = r_{t+1} + \gamma V(s_{t+1}) - V(s_t) \quad (2.2)$$

智能体的角色是根据由 θ 参数化的策略来选择动作。智能体利用来自评判者的反馈——特别是时序差分误差 δ_t ——来更新其策略参数。时序差分误差 δ_t 表示价值估计之间的时间差异，并为动作 a_t 相较于期望值究竟更优或更劣提供了信号（Schulman 等人, 2015b; Sutton 与 Barto, 2018）。智能体-评判器方法构成了近端策略优化（PPO）的基础，后者进一步优化策略更新，以确保稳定性并提升性能。

近端策略优化。近端策略优化（PPO）通过引入一种限制策略更新步长的方法，优化了

prevents large updates that can destabilize training (Schulman et al., 2017; Son et al., 2024). Specifically, PPO uses the probability ratio between the current policy π_θ and the old policy $\pi_{\theta_{\text{old}}}$:

$$r_t(\theta) = \frac{\pi_\theta(a_t|s_t)}{\pi_{\theta_{\text{old}}}(a_t|s_t)} \quad (2.3)$$

To constrain policy updates within a small range, PPO employs a clipped objective function. The clipped objective ensures that the probability ratio $r_t(\theta)$ remains within a specified range, typically $[1 - \epsilon, 1 + \epsilon]$, where ϵ is a small constant controlling the step size. The objective function for PPO is defined as:

$$J^{\text{CLIP}}(\theta) = \mathbb{E}_t [\min(r_t(\theta)A_t, \text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon)A_t)] \quad (2.4)$$

Here, $\text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon)$ limits the value of $r_t(\theta)$ to the range $[1 - \epsilon, 1 + \epsilon]$, thus preventing large policy updates that could destabilize training. The advantage function A_t represents the relative benefit of taking an action at a specific time step, compared to the expected value of following the policy from that state onward.

To further improve the efficiency of the advantage function estimation, PPO incorporates Generalized Advantage Estimation (GAE) (Schulman et al., 2015b). The advantage function A_t in PPO is computed using GAE as follows:

$$A_t = \sum_{k=0}^{T-t-1} (\gamma\lambda)^k \delta_{t+k} \quad (2.5)$$

where $\delta_{t+k} = r_{t+k+1} + \gamma V(s_{t+k+1}) - V(s_{t+k})$ is the TD error, and λ (a constant between 0 and 1) is used to adjust the weights of the TD errors over various time steps, acting as a trade-off hyperparameter. GAE allows PPO to efficiently approximate the advantage

Actor-Critic算法，从而避免可能导致训练不稳定的大幅更新（Schulman 等人，2017；

Son 等人，2024）。具体而言，PPO 利用当前策略 π_θ 与旧策略 $\pi_{\theta_{\text{old}}}$ 之间的概率比：

$$r_t(\theta) = \frac{\pi_\theta(a_t|s_t)}{\pi_{\theta_{\text{old}}}(a_t|s_t)} \quad (2.3)$$

为将策略更新限制在较小范围内，PPO采用裁剪目标函数。该裁剪目标函数可

确保概率比 $r_t(\theta)$ 始终保持在指定区间内——通常为 $[1 - E, 1 + E]$ ，其中 E 为控制

步长的微小常数。PPO的目标函数定义为：

$$J^{\text{CLIP}}(\theta) = \mathbb{E}_t[\min(r_t(\theta)A_t, \text{clip}(r_t(\theta), 1 - E, 1 + E)A_t)] \quad (2.4)$$

此处，裁剪项 $(r_t(\theta), 1 - E, 1 + E)$ 将 $r_t(\theta)$ 的取值限制在区间 $[1 - E,$

$1 + E]$ 内，从而避免可能导致训练失稳的大幅策略更新。优势函数 A_t 表示在特定

时间步采取某项动作的相对收益，相较于从此状态起遵循该策略的预期值。

为进一步提升优势函数估计的效率，PPO引入了广义优势估计（GAE）

（Schulman 等人，2015b）。PPO 中的优势函数 A_t 通过 GAE 按如下方式计算：

$$A_t = \sum_{k=0}^{T-t-1} (\gamma\lambda)^k \delta_{t+k} \quad (2.5)$$

其中 $\delta_{t+k} = r_{t+k+1} + \gamma V(s_{t+k+1}) - V(s_{t+k})$ 表示TD误差，而 λ （一个介于0到1之间的

常数）用于调节不同时间步长下TD误差的权重，充当权衡超参数。GAE通过累

加多个时间步长的折扣化TD误差，使PPO能够高效地近似优势函数 A_t 。裁剪目

function A_t by summing the discounted TD errors over multiple time steps. The clipped objective function $J^{\text{CLIP}}(\theta)$ helps in maintaining stable policy updates by ensuring that the updates do not deviate significantly from the previous policy. By integrating the clipped objective and GAE, PPO enhances the stability and performance of the Actor-Critic method. Compared to other reinforcement learning algorithms, such as Deep Q-learning (DQN) (Mnih et al., 2015), which is prone to overestimation bias and instability during training (Zha et al., 2021b), PPO offers a more stable approach due to its clipped objective and GAE mechanism. In this research, PPO is employed to improve the performance of the main-policy model in the Hierarchical Deep Reinforcement Learning (HDRL) framework for complex imperfect-information card games.

Advantages and Limitations. To provide a clear comparison, Table 2.1 summarizes the advantages and limitations of the above-discussed Reinforcement Learning methods.

Table 2.1: Advantages and Limitations of Reinforcement Learning Methods

Reinforcement Learning	Advantages	Limitations
Monte Carlo Method (Binder, 2022; Luengo et al., 2020)	1. Suitable for episodic tasks. 2. Can estimate state values or action values. 3. Can handle environments with unknown dynamics.	1. High variance in estimates. 2. Can be slow to converge. 3. May not work well in non-episodic or continuous tasks.
Temporal-Difference method (Ren et al., 2024)	1. Efficient and can update values after each time step. 2. Suitable for online learning. 3. Can handle both episodic and continuing tasks.	1. Limited exploration capability. 2. Can suffer from high bias in estimates. 3. Sensitive to initial conditions.
Actor-Critic (Padhye and Lakshmanan, 2023)	1. Combines advantages of both policy and value-based methods. 2. Provides stable learning and faster convergence. 3. Effective in continuous action spaces.	1. Requires tuning of hyperparameters. 2. Complex architecture and training. 3. Prone to overestimation bias.
Proximal Policy Optimization (PPO) (Son et al., 2024)	1. Achieves state-of-the-art performance in various domains. 2. Stable and sample-efficient. 3. Uses a simple objective function.	1. Sensitive to hyperparameter settings. 2. Can have slow convergence in some cases. 3. May not be the best choice for every problem.

标函数 $JCLIP(\theta)$ 通过确保参数更新不会与先前策略发生显著偏离，有助于维持策略更新的稳定性。通过将裁剪目标函数与GAE相结合，PPO提升了ActorCritic方法的稳定性与性能。相较于其他强化学习算法（例如Deep Q-learning (DQN) (Mnih et al., 2015)，该算法在训练过程中易出现高估偏差与不稳定性问题 (Zha et al., 2021b) ），PPO凭借其裁剪目标函数与GAE机制，提供了一种更为稳定的解决方案。在本研究中，PPO 被用于提升分层深度强化学习（HDRL）框架中主策略模型在复杂不完全信息卡牌游戏中的性能。

优势与局限性。为便于清晰对比，表2.1总结了上述强化学习方法的优势与局限性。

表 2.1：强化学习方法的优势与局限性

强化学习	优势	局限性
蒙特卡洛方法 (Binder, 2022; Luengo等, 2020)	1. 适用于阶段性任务。 2. 可估计状态值或动作值。 3. 可应对动态特性未知的环境。	1. 估计值方差较大。 2. 收敛速度可能较慢。 3. 在非间歇性或连续型任务中可能效果不佳。
时间差方法 (Ren 等人, 2024)	1. 高效，且可在每次操作后更新数值 时间步 2. 适用于在线学习。 3. 可同时处理单次任务和持续任务。	1. 探索能力有限。 2. 在估计值中可能出现高偏差。 3. 对初始条件敏感。
演员-评论家模型 (Padhye 与Lakshmanan, 2023)	1. 兼具政策与.....的优势 基于价值的方法。 2. 可提供稳定的学习效果及更快的收敛速度。 3. 在连续动作空间中有效。	1. 需要调整超参数。 2. 复杂的架构与训练。 3. 易产生高估倚倚。
近端策略优化 (PPO) (Son) 等人, 2024)	1. 实现了最先进的性能 在多个领域。 2. 稳定且样本效率高。 3. 采用一个简单的目标函数。	1. 对超参数设置较为敏感。 2. 在某些情况下，收敛速度可能较慢。 3. 可能并非最佳选择 针对每一个问题。

2.2.2 Hierarchical Deep Reinforcement Learning

Hierarchical Deep Reinforcement Learning (HDRL) builds on Deep Reinforcement Learning (DRL) by adding a structured hierarchy to better manage complex decision-making tasks (Chai et al., 2023; Gao et al., 2024; Vezhnevets et al., 2017). In HDRL, goals are divided into high-level and low-level sub-goals, each handled by different models, as shown in Figure 2.2. The high-level model inputs the state of the environment and outputs high-level actions or sub-goals based on its policy. These decisions are passed down to a low-level model, which executes the specific actions needed to achieve the sub-goals (Kulkarni et al., 2016; Ma et al., 2024b). For instance, a robot's high-level goal could be "make a sandwich," with sub-goals like "get bread," "get cheese," and "get ham," and each sub-goal having specific sub-policies such as "move to the fridge" → "open the fridge" → "take out cheese," all executed by the low-level model. The HDRL process has several steps: Initially, policies and value functions for both high-level and low-level models are trained using DRL algorithms like Proximal Policy Optimization (PPO) or Deep Q-learning (DQN). The high-level model observes the global state and selects sub-goals to maximize overall reward. These sub-goals are communicated to the low-level model, which perform the specific actions. The low-level model interacts with the environment, observing states and rewards. This reward is shared between high-level and low-level actions, providing feedback for both models. Policies are updated based on observed rewards and state transitions to enhance performance over time. An example of HDRL is the Hierarchical DQN (h-DQN) (Kulkarni et al., 2016; Zhao et al., 2023a), which combines hierarchical decision-making with the Deep Q-learning algorithm. This approach is effective in environments with sparse and delayed rewards, aligning short-term objectives with long-term goals for more efficient learning.

2.2.2 分层深度强化学习

分层深度强化学习 (HDRL) 在深度强化学习 (DRL) 的基础上, 通过引入结构化层次来更好地处理复杂的决策任务 (Chai等人, 2023; Gao等人, 2024; Vezhnevets等人, 2017)。在 HDRL 中, 目标被划分为高层子目标和低层子目标, 每个子目标由不同的模型处理, 如图2.2所示。高层模型接收环境状态, 并根据其策略输出高层动作或子目标。这些决策会被传递至低层模型, 由后者执行实现这些子目标所需的具体动作 (Kulkarni 等人, 2016; Ma 等人, 2024b)。例如, 机器人的高层目标可以是“制作三明治”, 其子目标包括“获取面包”、“获取奶酪”和“获取火腿”, 每个子目标又包含具体的子策略, 如“移至冰箱”→“打开冰箱”→“取出奶酪”, 所有这些操作均由低层模型执行。HDRL 过程包含多个步骤: 首先, 使用 DRL 算法 (如近端策略优化 (PPO) 或深度 Q 学习 (DQN)) 训练高层和底层模型的策略及价值函数。高层模型观察全局状态, 并选择子目标以最大化总体奖励。这些子目标被传递至底层模型, 由底层模型执行具体动作。底层模型与环境交互, 观察状态及奖励。这些奖励在高层与底层模型之间共享, 为两个模型提供反馈。策略根据观察到的奖励及状态转移进行更新, 从而逐步提升性能。HDRL 的一个典型应用是分层 DQN (h-DQN) (Kulkarni 等, 2016; Zhao 等人, 2023a), 该方法将分层决策与深度 Q 学习算法相结合。这种方法在奖励稀疏且延迟的环境中尤为有效, 能够将短期目标与长期目标相协调, 从而实现更高效的学习和更优的性能。受 HDRL 概念的启发, 本研究采用双层

and better performance. Inspired by HDRL concepts, a two-level HDRL framework is adopted to enhance performance. The framework integrates methods like Oracle Guiding, Adaptive Deep Monte Carlo (ADMC), Relative Advantage Reward Shaping (RARS), and PPO to improve the performance in complex imperfect-information card games such as DouDiZhu and Big2. ADCMC and PPO are distinct Reinforcement Learning methods that contribute to the hierarchical structure of HDRL.

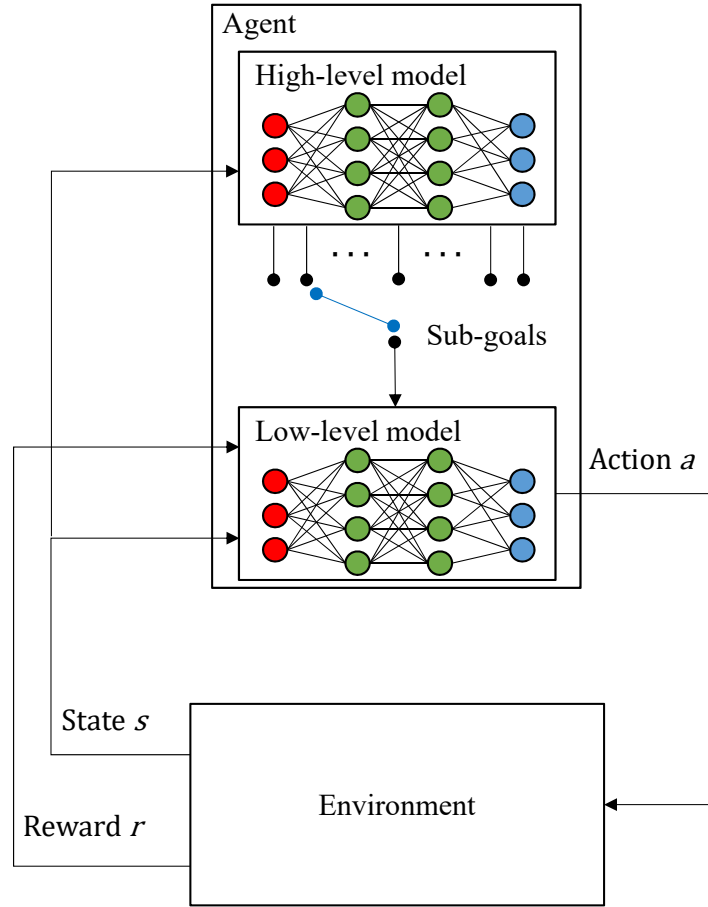


Figure 2.2: Hierarchical Deep Reinforcement Learning. Hierarchical Deep Reinforcement Learning (HDRL) divides decision-making into a high-level model that sets sub-goals and a low-level model that executes specific actions to achieve these sub-goals.

HDRL 框架以提升性能。该框架整合了多种方法，如 Oracle Guiding、自适应深度蒙特卡洛（ADMC）、相对优势奖励塑造（RARS）以及 PPO，旨在提高在复杂不完全信息卡牌游戏（如 DouDiZhu 和 Big2）中的表现。ADMC 与 PPO 是两种不同的强化学习方法，共同构成了 HDRL 的分层结构。

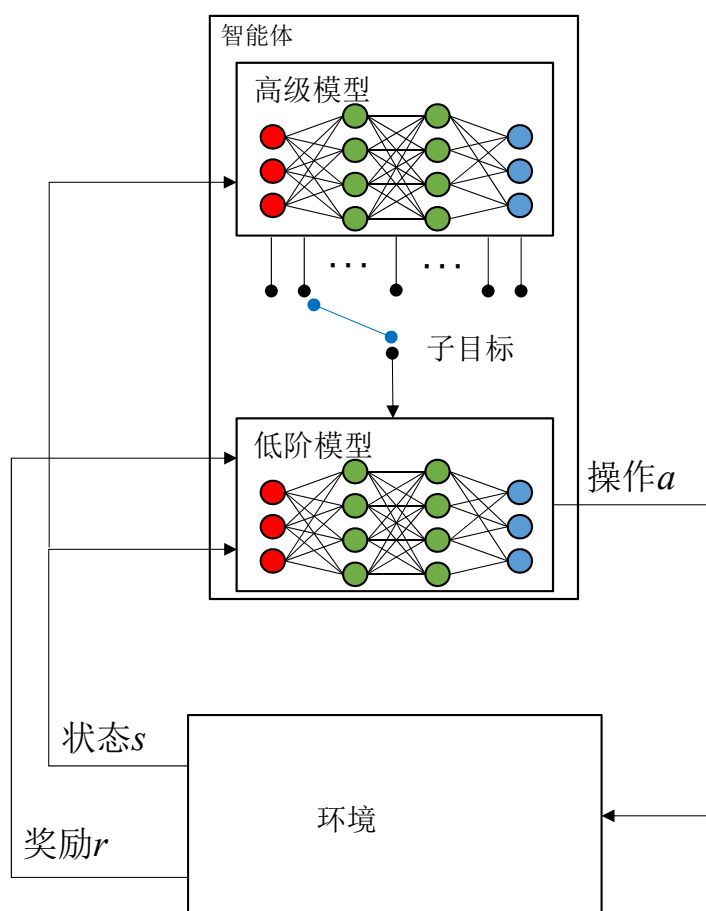


图 2.2: 分层深度强化学习。分层深度强化学习心智学习（HDRL）将决策过程划分为两个模型：一个高层模型负责设定子目标，另一个低层模型则执行具体行动以实现这些子目标。